



Low-temperature direct synthesis of layered m-LiMnO₂ for lithium-ion battery applications



Hui Zhou^a, Yingshun Li^a, Jiaolong Zhang^a, Wenpei Kang^b, Denis Y.W. Yu^{a, b, *}

^a School of Energy and Environment, City University of Hong Kong, Hong Kong, China

^b Center of Super-Diamond and Advanced Films (COSDAF), City University of Hong Kong, Hong Kong, China

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ABSTRACT

Layered monoclinic LiMnO₂ is a potential low-cost alternative to LiCoO₂ cathode for lithium-ion batteries. However, its application is limited partly because it cannot be synthesized easily. In this work, we successfully synthesized phase-pure layered LiMnO₂ in a one-step low-temperature process using electrolytic manganese dioxide (EMD) as the manganese precursor at 450 °C in Ar atmosphere by a carbothermal reduction method. The initial carbon to manganese ratio has to be optimized to remove impurities. The particle size of the material is between 40 and 60 nm. The synthesized sample shows an initial reversible capacity of about 180 mAh g⁻¹. The synthesis method is further applied to make boron-doped samples, which significantly enhanced the stability of the material, especially at high temperature. The boron-doped samples maintain a capacity of 150 mAh g⁻¹ for 100 cycles and give a capacity of about 100 mAh g⁻¹ at a current of 1000 mA g⁻¹ (equivalent to 10C rate). The ability to directly synthesize the material at low temperature is crucial because it allows further improvement of material performances through methods such as surface coating and the use of electrolyte additives to reduce surface reaction and Mn-dissolution.

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1. Introduction

Layered lithium transition metal compounds (LiMO₂) are excellent cathode material for lithium-ion batteries because the structure allows effective insertion and removal of lithium during charge and discharge [1,2]. Among the different materials, layered LiCoO₂ is commonly used in rechargeable lithium-ion batteries. However, the price of cobalt is high and volatile. With the increase in demand for large-scale applications such as electric vehicles and renewable energy systems, there is a need to develop alternative low-cost cathode materials.

Lithium manganese oxides are attractive choices as manganese is less toxic and inexpensive compared to cobalt [3–5]. Spinel LiMn₂O₄ is already being used as cathode material in some commercial cells, though its drawback is its low capacity of ~100 mAh g⁻¹ [6]. The available capacity of the manganese-based cathode material can be increased by using LiMnO₂ which contains a larger initial amount of lithium. LiMnO₂ crystallizes in two

phases: orthorhombic (o-LiMnO₂) with a zigzag arrangement of Li and Mn in the structure, and monoclinic (m-LiMnO₂) with a layered structure. Even though o-LiMnO₂ can be easily synthesized in inert atmosphere, it is difficult to be used for battery applications [7–10]. It requires an initial activation step to move the Li and Mn atoms within the lattice, which often results in a change in capacity with cycling [11–13]. On the other hand, m-LiMnO₂ has been demonstrated to give high capacity with good cycle performance with its layered structure, but the usage of m-LiMnO₂ has been limited partly because it cannot be synthesized easily [4,5,14,15]. Till now, m-LiMnO₂ was only synthesized through ion-exchange from NaMnO₂ [4,5] or high-temperature synthesis (typically >900 °C) with Al doping [16,17] or Cr doping [15,18]. These methods do not allow much control of the resulting product and also can lead to minor impurities in the material. In addition, Mn-dissolution and transformation to spinel are traditionally considered the Achilles' heels of the material.

The stability problems of m-LiMnO₂ originate from both the bulk and the surface of the material, and have to be solved collectively by stabilizing the bulk structure with dopants and protecting the surface with coating to prevent Mn-dissolution and surface reactions. Attempts were made mainly in the 1990s. But since then, there have been many advances in material synthesis and

* Corresponding author. School of Energy and Environment, City University of Hong Kong, Hong Kong, China.

E-mail address: denisyu@cityu.edu.hk (D.Y.W. Yu).

preparation techniques. For instance, new electrolytes are more pure, as well as more resistant to decomposition. Surfaces are more stable against side reactions with the help of surface coating techniques such as atomic layer decomposition (ALD). Morphology, particle size and structure of the materials are more well-defined with new chemical synthesis methods. All the ingredients that are necessary to tackle the issues of m-LiMnO₂ are ready. So the first step to re-visit the m-LiMnO₂ material for battery applications is to find a simple method to synthesize it so that material modifications can be applied.

In this paper, we report for the first time a direct low-temperature synthesis of layered m-LiMnO₂ through carbo-thermal reduction with LiOH and MnO₂ as precursors under Ar atmosphere. The layered material can be made easily with the low-temperature synthesis, which allows better control of the structure, composition and particle size of the material, which are essential for further improvement of its properties. Phase-pure m-LiMnO₂ is synthesized through careful optimization of the type of MnO₂ precursor and the C/Mn ratio. Discharge capacity of 180 mAh g⁻¹ is obtained from the as-made m-LiMnO₂. We then apply the same synthesis process to dope the material with different elements to improve material stability. B-doping has so far been the most successful in stabilizing the cycle performance of m-LiMnO₂, especially at elevated temperature, and the results are introduced here. The work opens the door for future work on surface coating of the material to further stabilize the materials against Mn-dissolution and phase transition.

2. Material and methods

2.1. Materials preparation

Stoichiometric ratio (1:1) of LiOH·H₂O and electrolytic manganese dioxide (EMD) or β-MnO₂ (Alfa Aesar) were mixed with different amount of acetylene black (Alfa Aesar) by ball milling (200 rpm with stainless steel can and balls) for 1 h with acetone. About 1.5 g mixed precursor was pressed into a pellet with a force of 3 tons and annealed in a furnace (R. D. WEBB) under Ar gas flow at 450 °C or 600 °C for 15 h with a heating rate of 10 °C/min. The same synthesis procedure is used for B-doped samples, with the addition of H₃BO₃ in the precursor.

2.2. Material characterizations

The structure of the samples was characterized by X-ray powder diffraction (XRD) using a BRUKER diffractometer (AXS D2 Phaser) equipped with Cu K_α source (λ = 1.54178 Å). Data were collected over a 2θ range of 10–80° with a step size of 0.02° and dwell time of 0.1 s. Scanning electron microscopy (SEM) was carried out using a Philips XL30 FEG SEM. The samples were dispersed on a Si wafer for the SEM measurements. Transmission electron microscopy (TEM) was carried out for the microstructural investigation of the samples using a Tecnai F20 microscope operating at 200 kV. Thermogravimetric analysis (TGA) was performed under air or N₂ atmosphere at a heating rate of 10 °C/min from room temperature to 1000 °C with a TGA/DSC 1 (METTLER TOLEDO) instrument. Elemental analysis was performed by a vario EL III Element Analyzer. Generally, 3 repeated tests were done for each sample and the averaged values are reported.

2.3. Electrochemical characterizations

The electrochemical properties were evaluated in two-electrode coin-type cells (size 2032) on a Neware Battery Testing System using lithium foil as the counter electrode. A slurry for each sample

was prepared by mixing the sample powder, acetylene black, and polyvinylidene fluoride (PVDF) with a weight ratio of 80:10:10 in 1-methyl-2-pyrrolidone (NMP). The homogeneous slurry was then tape-casted on Al foil with a doctor blade and then dried on a hotplate at 80 °C. After roll pressing and punching, the electrodes were vacuum-dried for 4 h at 110 °C, then put in the Ar-filled glove-box for coin cell assembly. Mass loading of the cathode was around 2 mg/cm² with an electrode area of 2.0 cm². The electrolyte was 1.0 M lithium hexafluorophosphate (LiPF₆) dissolved in a mixture of ethylene carbonate (EC) and diethyl carbonate (DEC) with a volume ratio of 1:1. Celgard 2325 separators (Celgard Inc., USA) were used. The coin cells were assembled in a high-purity argon-filled glove box. The charge–discharge cycling (at 25 °C) was performed between 2.0 and 4.5 V with a current density of 50 mA g⁻¹ or higher for rate test.

3. Results and discussion

Under usual conditions, such as a direct reaction between LiOH and Mn₂O₃ in Ar atmosphere, orthorhombic o-LiMnO₂ phase is obtained at a temperature above 600 °C, while a mixture of orthorhombic and monoclinic LiMnO₂ is obtained below that (see Fig. 1). Even though single-phase m-LiMnO₂ cannot be formed using trivalent Mn₂O₃ as the precursor, the result suggests that m-LiMnO₂ can be stable under a low-temperature synthesis with a proper choice of precursor and synthesis condition. Carbothermal reduction using tetravalent MnO₂ precursor is therefore attempted in this study.

Two different MnO₂ precursors were used as the precursor: electrolytic manganese dioxide (EMD) with predominantly γ-MnO₂, and β-MnO₂ with a tetragonal phase. Their XRD patterns are shown in Fig. 2(a). The crystal structures of these two MnO₂ are compared in Fig. 2(b): both of them have a tunnel structure consisting of interconnected MnO₆ octahedra, but 1 × 1 tunnel for β-MnO₂ and 1 × 2 tunnel for γ-MnO₂. MnO₂ is known to decompose to Mn₂O₃ and Mn₃O₄ at high temperature [19], and TGA was performed with N₂ gas flow on the two MnO₂ to determine the decomposition temperature. As shown in Fig. 2(c), decomposition of β-MnO₂ occurs in two steps, first at around 550 °C with a 9% drop in weight, and another at ~800 °C with an additional 4% drop in weight. This is consistent with the formation of Mn₂O₃ and Mn₃O₄, respectively. For EMD, weight loss starts gradually from 100 °C and continues until further decompositions at 450 °C and 750 °C. The initial weight loss at low temperature is associated with the removal of free and/or bound H₂O in EMD, which contains about 5wt% of water.

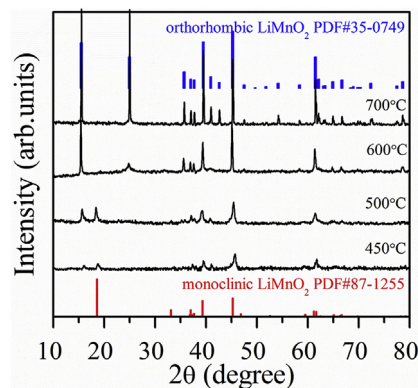


Fig. 1. XRD patterns of synthesized Li–Mn–O samples in Ar atmosphere with Mn₂O₃ as precursor at different temperatures.

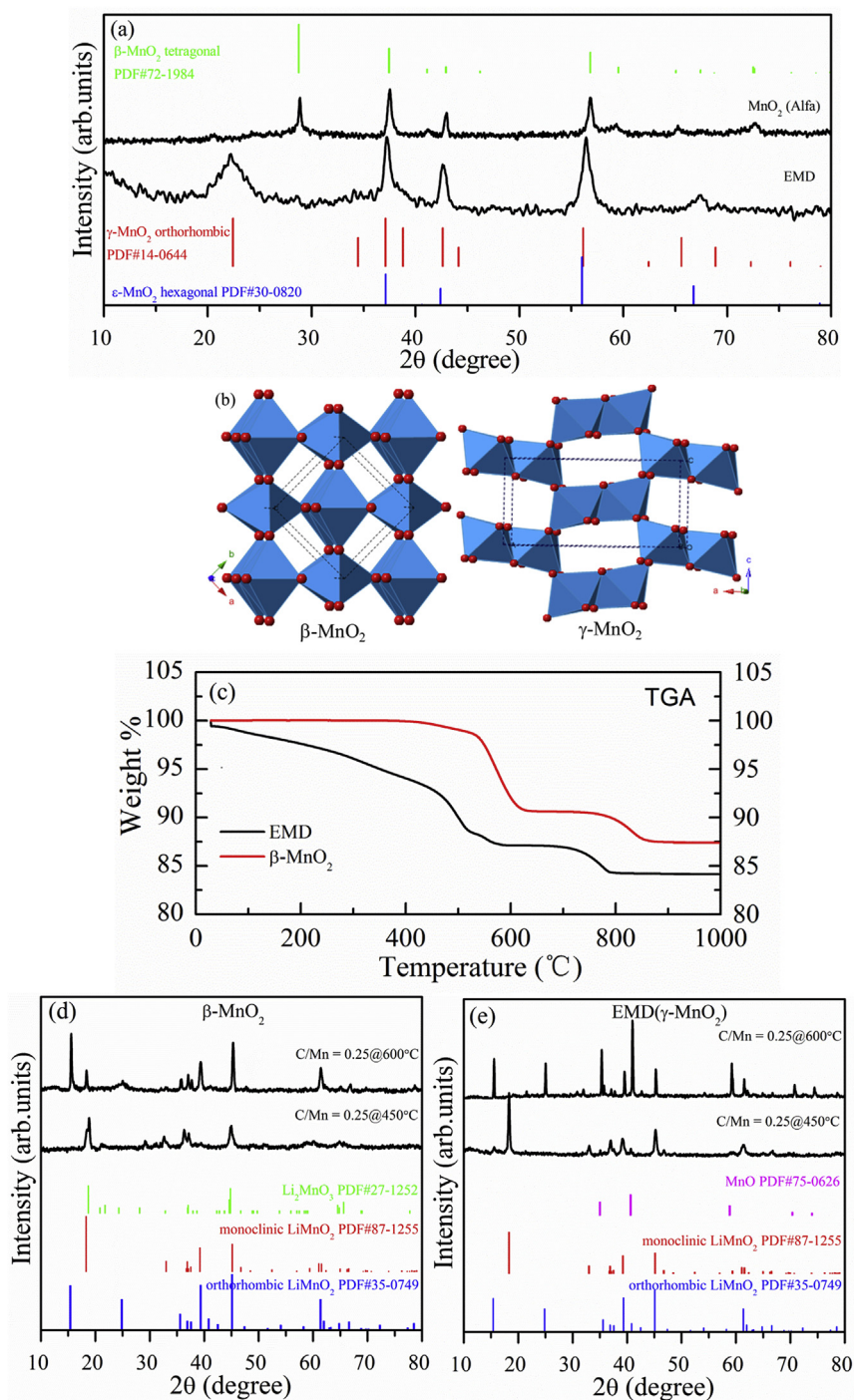
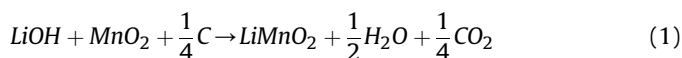


Fig. 2. (a) XRD patterns of EMD and β -MnO₂. (b) Crystal images for β -MnO₂ and γ -MnO₂. (c) TGA for the EMD and β -MnO₂ performed in N₂ atmosphere. (d) XRD patterns of the products synthesized with β -MnO₂ at 450 and 600 °C in Ar atmosphere. (e) XRD patterns of the products synthesized with EMD at 450 and 600 °C in Ar atmosphere.

Based on the TGA results, the annealing temperature for the synthesis of m-LiMnO₂ is chosen as 450 °C and 600 °C. Acetylene black is used as the reducing agent in the carbothermal reduction reaction. The reaction can be expressed as follows:



A complete reduction of Mn⁴⁺ to Mn³⁺ in theory requires 0.25 mol of carbon per mole of Mn. Precursors of LiOH, MnO₂ and

acetylene black were mixed by ball milling in stoichiometric ratios of Li:Mn:C = 1:1:0.25 using the two different MnO₂ precursors and annealed in Ar atmosphere. XRD results of the synthesized products are respectively shown in Fig. 2(d) and (e). Both samples annealed at 600 °C shows orthorhombic LiMnO₂ as the predominant phase in the materials. When the annealing temperature is reduced to 450 °C, sample with β -MnO₂ precursor shows mixed phases of Li–Mn–O with tetravalent oxide phases, whereas the sample with EMD shows predominantly monoclinic LiMnO₂. The results indicate that monoclinic LiMnO₂ can be synthesized with

EMD at low temperature. It is possible that the structural similarity between the γ - MnO_2 and m-LiMnO_2 help to obtain the m-LiMnO_2 .

For the sample with EMD synthesized at 450 °C, two minor peaks belonging to MnO are observed, suggesting the sample is over-reduced under this synthesis condition, i.e. the amount of carbon is in excess. To reduce the amount of MnO impurities and determine the appropriate amount of carbon needed for the carbothermal reaction, syntheses were repeated at 450 °C with reduced amount of carbon ($\text{C/Mn} = 0.125, 0.1, 0.0833, 0.0625$). XRD patterns of the synthesized samples are compared in Fig. 3(a). Phase-pure monoclinic LiMnO_2 is obtained when the C/Mn ratio is between 0.1 and 0.0833. When the carbon content is reduced to 0.0625, impurity phases corresponding to Li_2MnO_3 and other lithium manganese oxide phases with Mn^{4+} are observed, indicating insufficient reduction with lower carbon content. The

calculated lattice parameters for the $\text{C/Mn} = 0.0833$ sample is $a = 5.438 \text{ \AA}$, $b = 2.800 \text{ \AA}$, $c = 5.382 \text{ \AA}$, $\beta = 116.2^\circ$ with a C2/m space group, which is close to the reported value for monoclinic LiMnO_2 : $a = 5.439 \text{ \AA}$, $b = 2.808 \text{ \AA}$, $c = 5.388 \text{ \AA}$, $\beta = 116.0^\circ$ [4].

The successful synthesis of phase-pure monoclinic LiMnO_2 requires less carbon than in Eq. (1). This is attributed to the self-decomposition of EMD, as shown in Fig. 2(c). Elemental analyses also show carbon residues after annealing in the samples with C/Mn of 0.125 and 0.25 are 1.0wt% and 2.7wt%, respectively, while the residue is nearly zero with C/Mn between 0.1 and 0.0833. This further confirms that less carbon is needed for the carbothermal reaction.

The samples with different C/Mn ratios were characterized by SEM. SEM pictures (Fig. 3(b)) show that the appearance of the resulting LiMnO_2 after synthesis is similar to the starting EMD

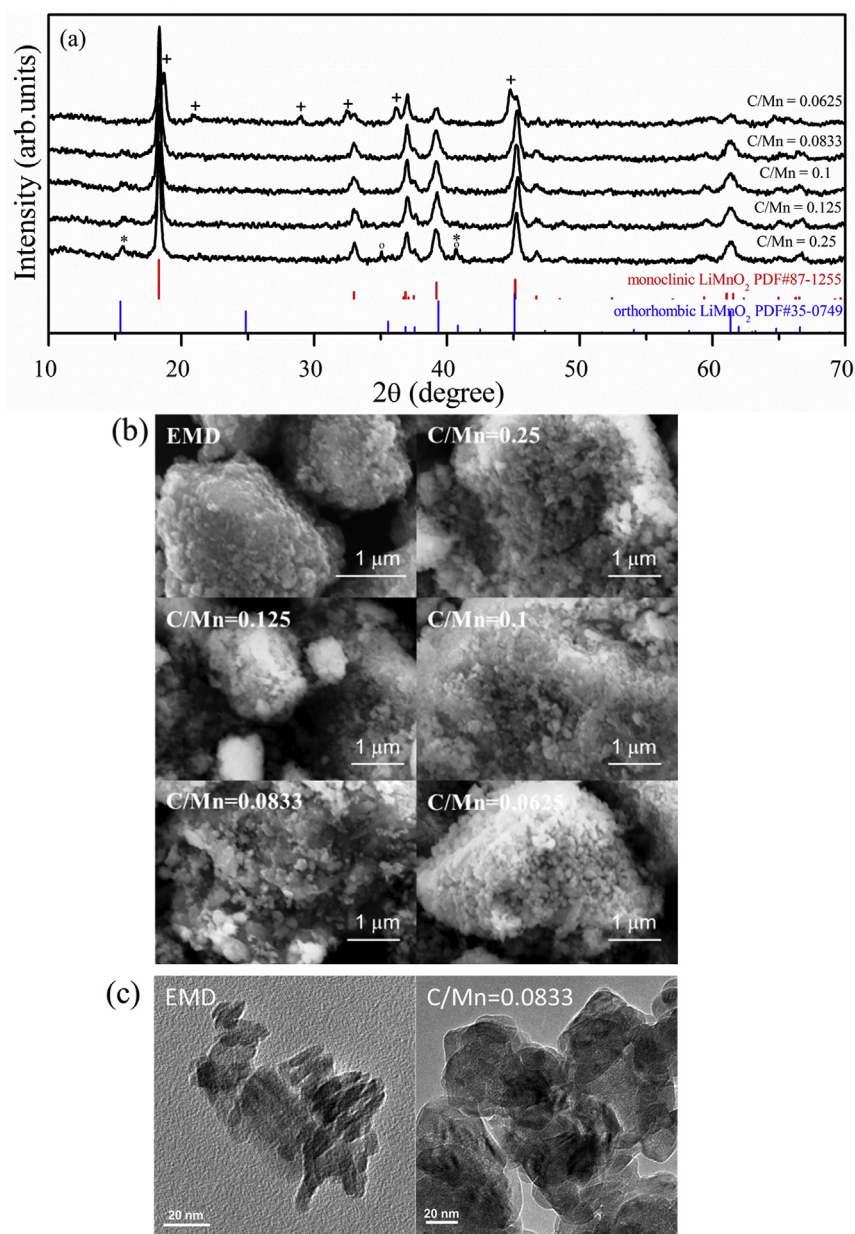


Fig. 3. (a) XRD patterns of synthesized LiMnO_2 samples using EMD as precursor with different C/Mn ratios (impurities are indicated by symbols in the figure, *: orthorhombic LiMnO_2 ; o: MnO ; +: Li_2MnO_3 and other tetraivalent lithium manganese oxide). (b) SEM pictures of the EMD precursor and the synthesized LiMnO_2 samples with different C/Mn ratios. (c) TEM pictures of the EMD precursor and the synthesized monoclinic LiMnO_2 sample ($\text{C/Mn} = 0.0833$).

precursor. This suggests that the morphology and particle size of the m-LiMnO₂ product can be easily manipulated by adjusting the Mn precursor. This is crucial for future development of the materials. TEM images (Fig. 3(c)) show that the EMD precursor is nano-sized and the resulting m-LiMnO₂ has a particle size between 40 and 60 nm. The low-temperature synthesis process prevents significant grain growth of the material.

The electrochemical performance of the synthesized monoclinic LiMnO₂ samples with different C/Mn ratios were evaluated and compared in Fig. 4(a). The phase-pure sample (C/Mn = 0.0833) shows the highest capacity and best cycle performance: the initial reversible capacity is about 180 mAh g⁻¹ and a capacity of above 150 mAh g⁻¹ is still maintained after 50 cycles. The charge–discharge curves of the C/Mn = 0.125 and 0.0833 samples (Fig. 4(b) and (c)) clearly show the development of 3 V/4 V plateaus with cycling, suggesting a layered-to-spinel transformation with lithium extraction and insertion, which is expected for single-phase non-doped m-LiMnO₂ without surface protections. The initial C/Mn ratio shows a large effect on the reversibility of the “3 V” plateau, which may be due to non-stoichiometry of the product with different C/Mn ratios.

Even though the capacity of the as-synthesized m-LiMnO₂ start to drop after 30 cycles, the direct synthesis method we have developed is an excellent starting point for further development of

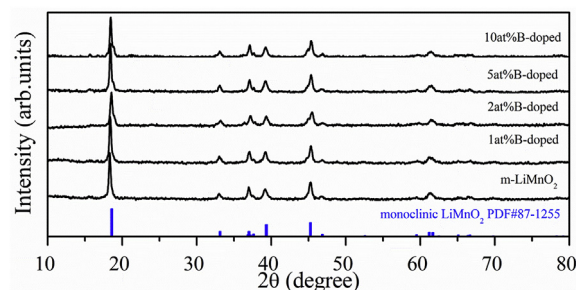


Fig. 5. XRD patterns of synthesized m-LiB_xMn_{1-x}O₂ samples with different dopant levels: 1 at% B, 2 at% B, 5 at% B and 10 at% B.

layered LiMnO₂. One method to stabilize the structure is to dope it with different elements. We have tried synthesizing doped materials with various dopants using the same preparation process. Among the different dopants we have tried, boron-doping so far gives the best material stability and the results are shown below.

Materials with 1 at% B (B:Mn = 1:99), 2 at% B (B:Mn = 2:98), 5 at% B (B:Mn = 5:95) and 10 at% B (B:Mn = 10:90) were synthesized with the same procedure with C/(B + Mn) ratio of 0.0833. No impurity is observed in any of the samples, as shown by XRD patterns

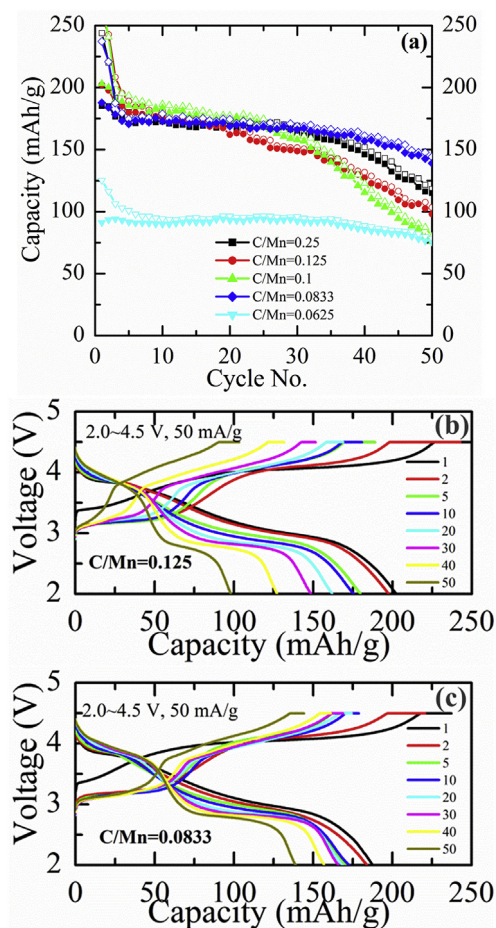


Fig. 4. (a) Cycling performance of the synthesized monoclinic LiMnO₂ samples with EMD as Mn precursor and different C/Mn ratios. (b) Charge–discharge curves at different cycle numbers of the synthesized m-LiMnO₂ sample (C/Mn = 0.125). (c) Charge–discharge curves at different cycle numbers of the synthesized m-LiMnO₂ sample (C/Mn = 0.0833).

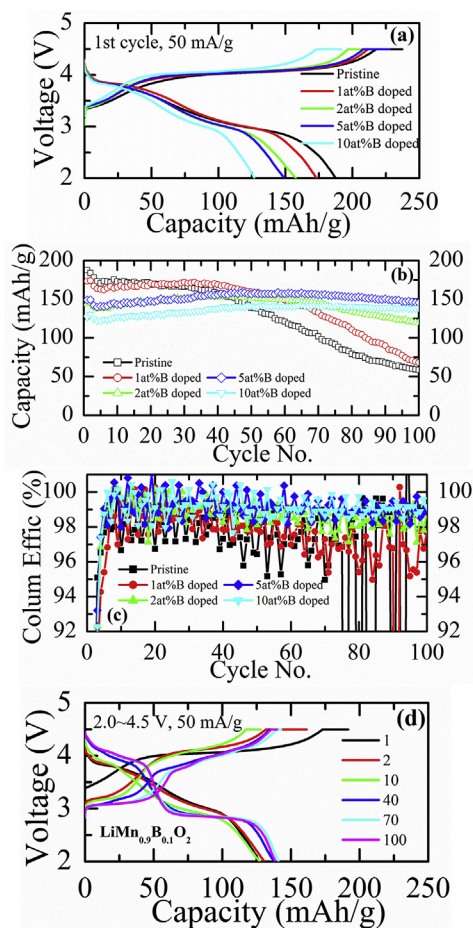


Fig. 6. (a) 1st charge–discharge curves of the synthesized m-LiB_xMn_{1-x}O₂ samples (C/Mn = 0.0833) with different B contents. (b) Cycle performance (at 25 °C) of m-LiB_xMn_{1-x}O₂ samples (C/Mn = 0.0833) with different B contents. (c) Coulombic efficiency of m-LiB_xMn_{1-x}O₂ samples (C/Mn = 0.0833) with different B contents. (d) Charge–discharge curves of the m-LiB_xMn_{1-x}O₂ sample (C/Mn = 0.0833) with 10 at% B-doping at different cycle numbers.

Table 1

Comparison of the averaged coulombic efficiency for B-doped m-LiMnO₂ samples (C/Mn = 0.0833).

Samples	Averaged coulombic efficiency at 25 °C	Averaged coulombic efficiency at 60 °C
Pristine	97.6%	69.7%
1 at% B doped	98.0%	99.1%
2 at% B doped	98.9%	99.1%
5 at% B doped	99.3%	99.1%
10 at% B doped	99.2%	99.0%

in Fig. 5. The XRD peaks shift slightly to larger angles with increasing B content, indicating smaller lattice parameters with B-incorporation, which is consistent with the smaller ionic radius of B than Mn.

The electrochemical performances of the B-doped samples were tested and the results are summarized below:

1. Charge-discharge capacity is reduced with B-doping (Fig. 6(a)). This is reasonable as boron is an electrochemically inactive element.
2. Cycle performance and coulombic efficiency are improved with B-doping (Fig. 6(b) and (c)).
3. Charge-discharge curves of B-doped samples show the development of both 3 V and 4 V plateaus with cycling (An example of the charge–discharge curves of m-LiB_{0.1}Mn_{0.9}O₂ is shown in Fig. 6(d)).

The results indicate that even though B-doping cannot ultimately suppress the layered-to-spinel transformation with cycling, coulombic efficiency is improved, leading to better cycle stability. Non-doped m-LiMnO₂ gives an average coulombic efficiency of about 97.6% (Table 1), probably due to electrolyte decomposition and/or transition metal dissolution. 5 at% B-doping increases coulombic efficiency to 99.2%. The beneficial effect of B is especially significant when the material is tested at elevated temperature (60 °C). Non-doped sample shows a coulombic efficiency of about 70% and the material fails after 20–30 cycles (see Fig. 7 and Table 1). The low coulombic efficiency and poor cycle stability most likely are due to accelerated surface reaction and/or Mn dissolution at the elevated temperature. High-temperature cycle performance is improved by B-doping, and coulombic efficiency is increased to about 99%. The nano-sized particles may contribute partially to the material stability by reducing the stress build-up due to the Jahn-Teller distortion, but it is not the main reason for the improved performance of the B-doped sample, as the non-doped sample is also nano-sized.

Among the tested samples, m-LiMnO₂ with 5 at% B gives the best compromise between capacity and cycle performance. Our synthesized m-LiMnO₂ shows higher capacity and better cycle stability than spinel LiMn₂O₄ and o-LiMnO₂ tested with the same voltage range (2.0–4.5 V) (Fig. 8(a)). Furthermore, m-LiMnO₂

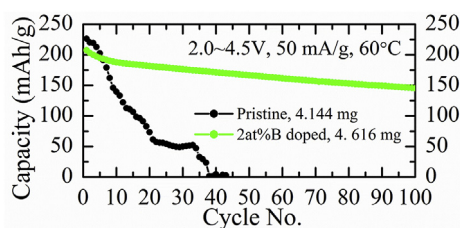


Fig. 7. Cycle performance at 60 °C of m-LiMnO₂ sample with or without B-doping (C/Mn = 0.0833).

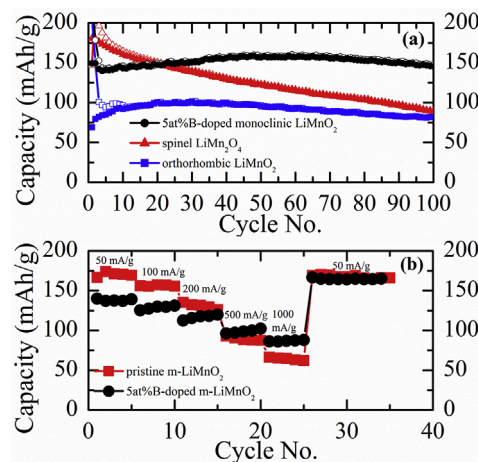


Fig. 8. (a) Cycle performance of B-doped m-LiMnO₂ (C/Mn = 0.0833), spinel LiMn₂O₄ and orthorhombic LiMnO₂, cycled between 2.0 and 4.5 V vs. Li/Li⁺ with a current of 50 mA g⁻¹. (b) Rate performance of m-LiMnO₂ sample (C/Mn = 0.0833).

sample shows good rate performance (Fig. 8(b)). A capacity of ~100 mAh g⁻¹ is obtained at a current of 1000 mA g⁻¹ (equivalent to 10C rate). Further work is necessary to further suppress side reactions and Mn-dissolution with surface coating or the use of electrolyte additives.

In summary, layered LiMnO₂ with controlled particle size and morphology can now be directly synthesized through our low-temperature carbothermal method. This opens the door for many more works in the area in search for a stable low-cost cathode material in the future.

4. Conclusion

We first report the direct synthesis of phase-pure monoclinic LiMnO₂ at low temperature using a carbothermal reduction method. The following conditions are needed: 1) a low-temperature synthesis to prevent the formation of orthorhombic phase; 2) an appropriate tetravalent Mn precursor that can react at low temperature; 3) a controlled reduction of the tetravalent Mn precursor with optimal C/Mn ratio to remove impurities. A high capacity of about 180 mAh g⁻¹ is obtained from the sample, and the capacity remains above 150 mAh g⁻¹ for at least 50 cycles. This process allows easy synthesis of the layered LiMnO₂ and opens up many future possibilities of the material. We have applied the method to synthesize B-doped m-LiMnO₂, which gives improved cycle performance, especially at high temperature, by reducing surface reaction and/or transition metal dissolution. The resulting B-doped m-LiMnO₂ material shows better performance than spinel LiMn₂O₄ and o-LiMnO₂. With further work on doping and surface protection, layered LiMnO₂ may become an attractive cathode material for future applications.

Authors' contribution

HZ carried out most of the materials synthesis and electrochemical tests. YL, JZ and WK performed the material characterizations. DYWY supervised the project.

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